

MAIN CONTROLLER BOARD

SAPPHIRE - REV A



RESPONSIBLE ENGINEER:
NICHOLAS ALVES

REVIEWERS:

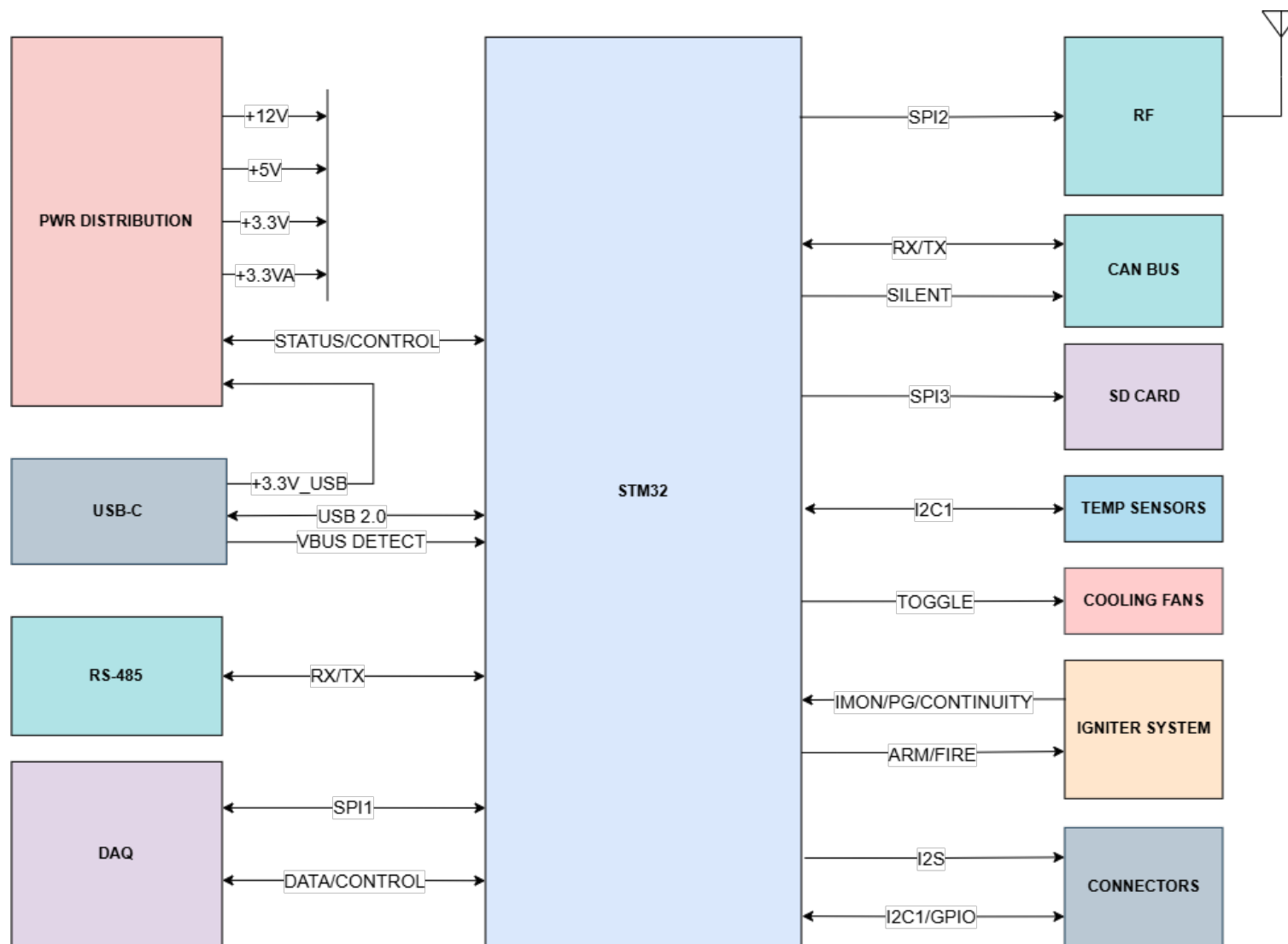
SCRT
STUDENT COMPETITION ROCKETRY TEAM

Sheet: /Title Page/
File: title_page.kicad_sch

Title: TITLE PAGE

Size: A4 Date: 2026-08-27
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Rev: A
Id: 1/27



Sheet: /Block Diagram/
File: block_diagram.kicad_sch

Title: BLOCK DIAGRAM

Size: A4 Date: 2026-08-27

KiCad E.D.A. 10.0.5

Rev: A

Id: 2/27

- Main GSE Board Pad Side:
- STM32 MCU
 - SD card & USB-C
 - 2x Temp sensors
 - 4x 4-wire PT/LC amps
 - 4x TC amps
 - CAN Bus for external DAQ comm
 - RS-485 for user-side comms
 - LoRa for user-side comms
 - 12V, 18A igniter
 - 2x 12V fan connectors
 - External valve control
 - External audio DAC output

power_distribution

5V_INTF_EN
3V3_INTF_EN
3V3_SD_EN
3V3_RADIO_EN
BAT_DISABLE
DC_JACK_DISABLE
JACK_DETECT
3V3_STATUS
12V_STATUS
3V3_MAIN_PG00DD
EFUSE_PG00DD
VIN_ADC
VIN_RAW_ADC
EFUSE_IMOND

File: power_distribution.kicad_sch
RS_485

{RS485_MAIN}

File: RS_485.kicad_sch
can_bus

TX
RXD
SILENT

File: can_bus.kicad_sch
pcb_sensors

SENS_SDA
SENS_SCL
TEMP1_ALERT
TEMP2_ALERT
ADC_RDY

File: pcb_sensors.kicad_sch
DAQ

SPI_MISO
SPI_MOSI
SPI_SCK
T1_CS
TC1_DRDY
TC1_FAULT
T2_CS
TC2_DRDY
TC2_FAULT
T3_CS
TC3_DRDY
TC3_FAULT
T4_CS
TC4_DRDY
TC4_FAULT
PT1_DOUT
PT1_SCK
PT2_DOUT
PT2_SCK
LC1_DOUT
LC1_SCK
LC2_DOUT
LC2_SCK
PT1_3WIRE_OUT
PT2_3WIRE_OUT

File: daq.kicad_sch

stm32_controller

5V_INTF_EN
3V3_INTF_EN
3V3_SD_EN
3V3_RADIO_EN
BAT_DISABLE
DC_JACK_DISABLE
JACK_DETECT
3V3_STATUS
STATUS_12V
3V3_MAIN_PG00DD
EFUSE_PG00DD
VIN_ADC
VIN_RAW_ADC
EFUSE_IMON

{RS485_MAIN}

CAN1_TX
CAN1_RX
CAN_SILENT

I2C1_SDA
I2C1_SCL
I2C1_ALERT
I2C2_ALERT
ADC_RDY

SPI1_MISO
SPI1_MOSI
SPI1_CLK
TC1_CS
TC1_DRDY
TC1_FAULT
TC2_CS
TC2_DRDY
TC2_FAULT
TC3_CS
TC3_DRDY
TC3_FAULT
TC4_CS
TC4_DRDY
TC4_FAULT
DAQ_DOUT1
DAQ_SCK1
DAQ_DOUT2
DAQ_SCK2
DAQ_DOUT3
DAQ_SCK3
DAQ_DOUT4
DAQ_SCK4
PT_3WIRE_IN1
PT_3WIRE_IN2

File: stm32_controller.kicad_sch

fans

FAN1_TOGGLED
FAN2_TOGGLED

File: fans.kicad_sch

valve & audio connectors

{IDC}
I2C1_SCL
I2C1_SDA
{INTF}

LRCK
SCLK
DIN
BCK

File: misc.kicad_sch

SD_card

{SD_CARD}

File: sd_card.kicad_sch

rf

{RADIO}

File: rf.kicad_sch

igniter_system

IGNITER_FIRED
CONTINUITY_SENSE
SHUNT_DISABLED
IGNITER_IMON
IGNITER_PG00DD

IGNITER_FIRE
CONTINUITY_SENSE
SHUNT_DISABLE
IMON
PGOOD_IGNITER

File: igniter_system.kicad_sch

USB_C

USB_VBUS_DET
USB_P
USB_N

USB_VBUS_DET
D_P
D_N

File: usb_c.kicad_sch



Sheet: /top-level/
File: master_gse_pcb_project.kicad_sch

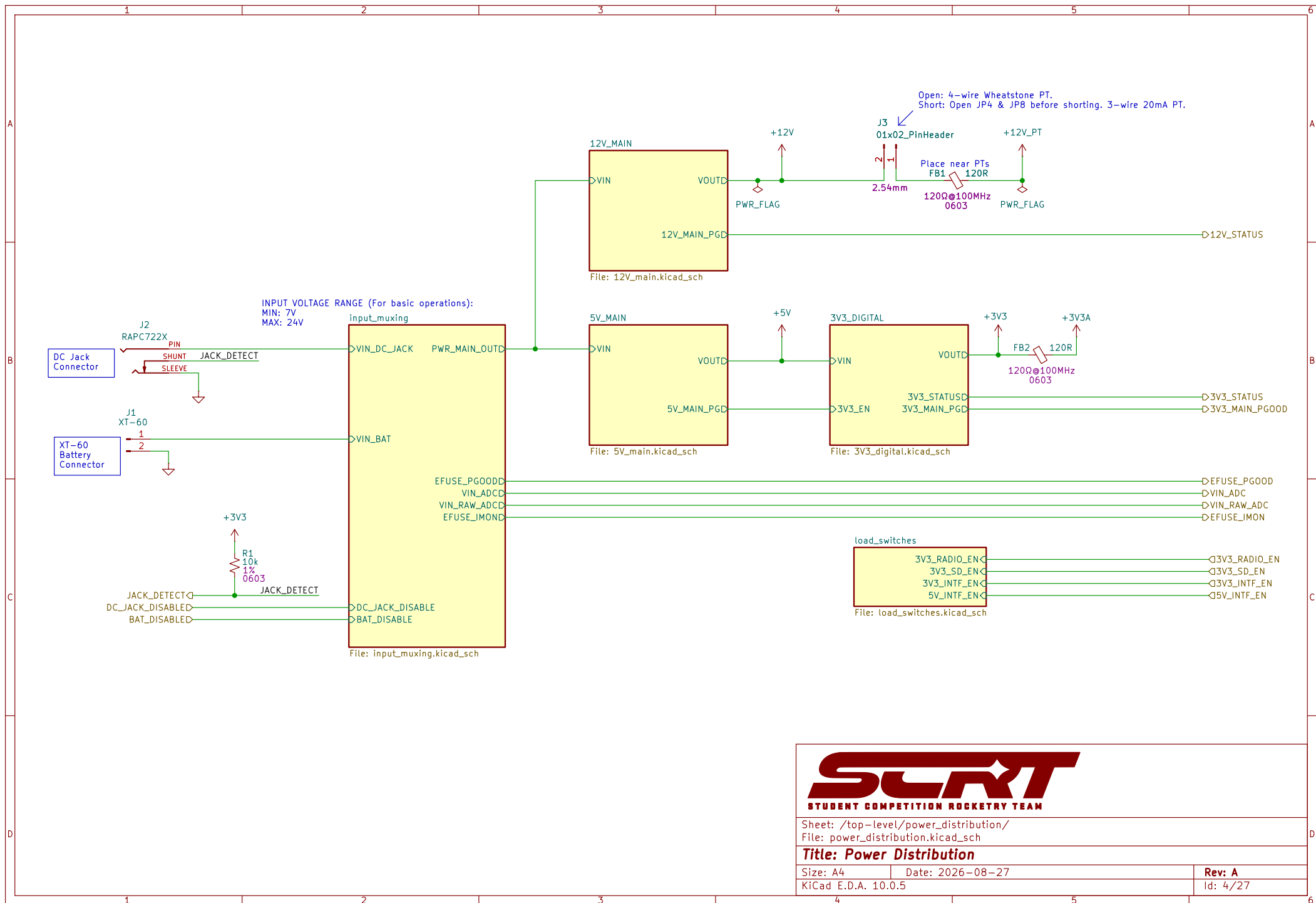
Title: Top Level

Size: A4 Date: 2026-08-27

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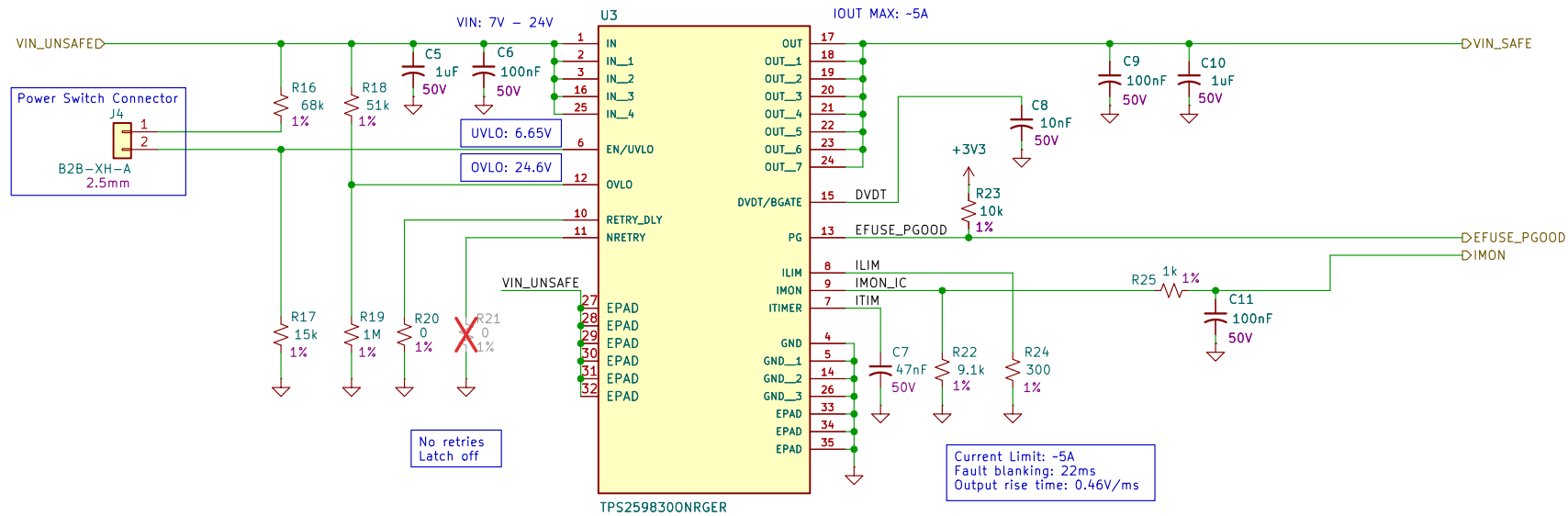
STUDENT COMPETITION ROCKETRY TEAM

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File: power_distribution.kicad_sch

Title: Power Distribution

Size: A4	Date: 2026-08-27	Rev: A
KiCad E.D.A. 10.0.5		Id: 4/27





Sheet: /top-level/power_distribution/input_muxing/efuse/
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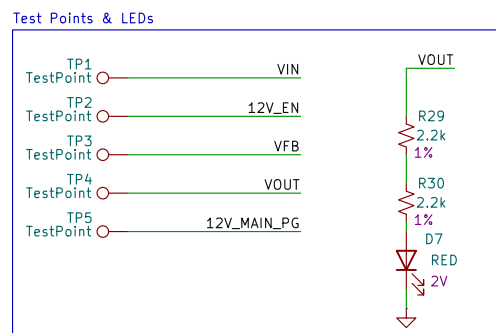
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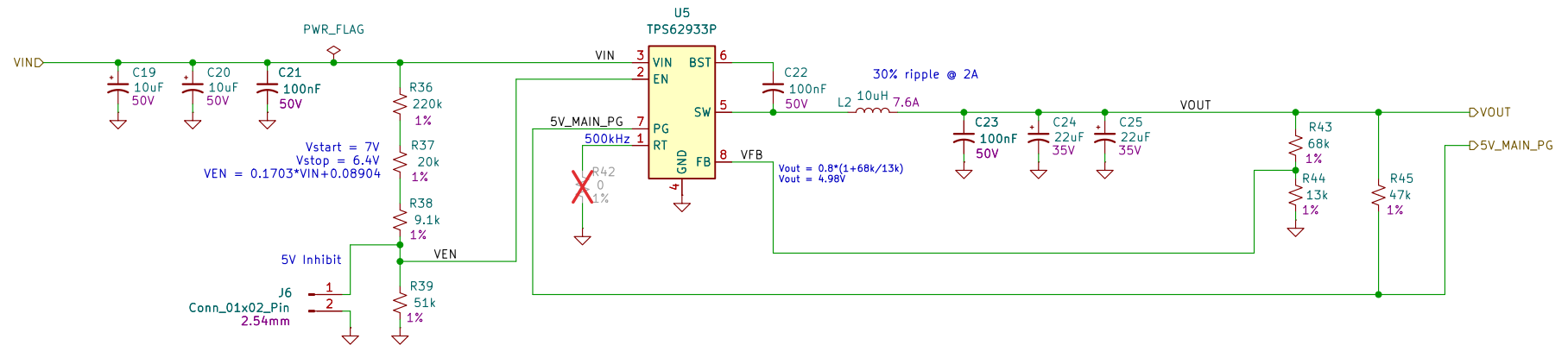
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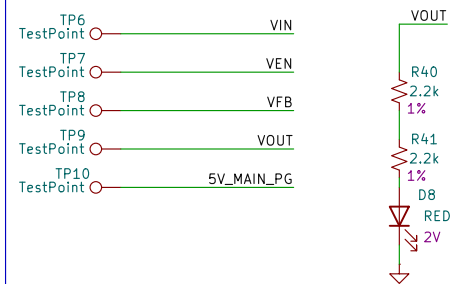
Rev: A

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Test Points & LEDs



Sheet: /top-level/power_distribution/5V_MAIN/
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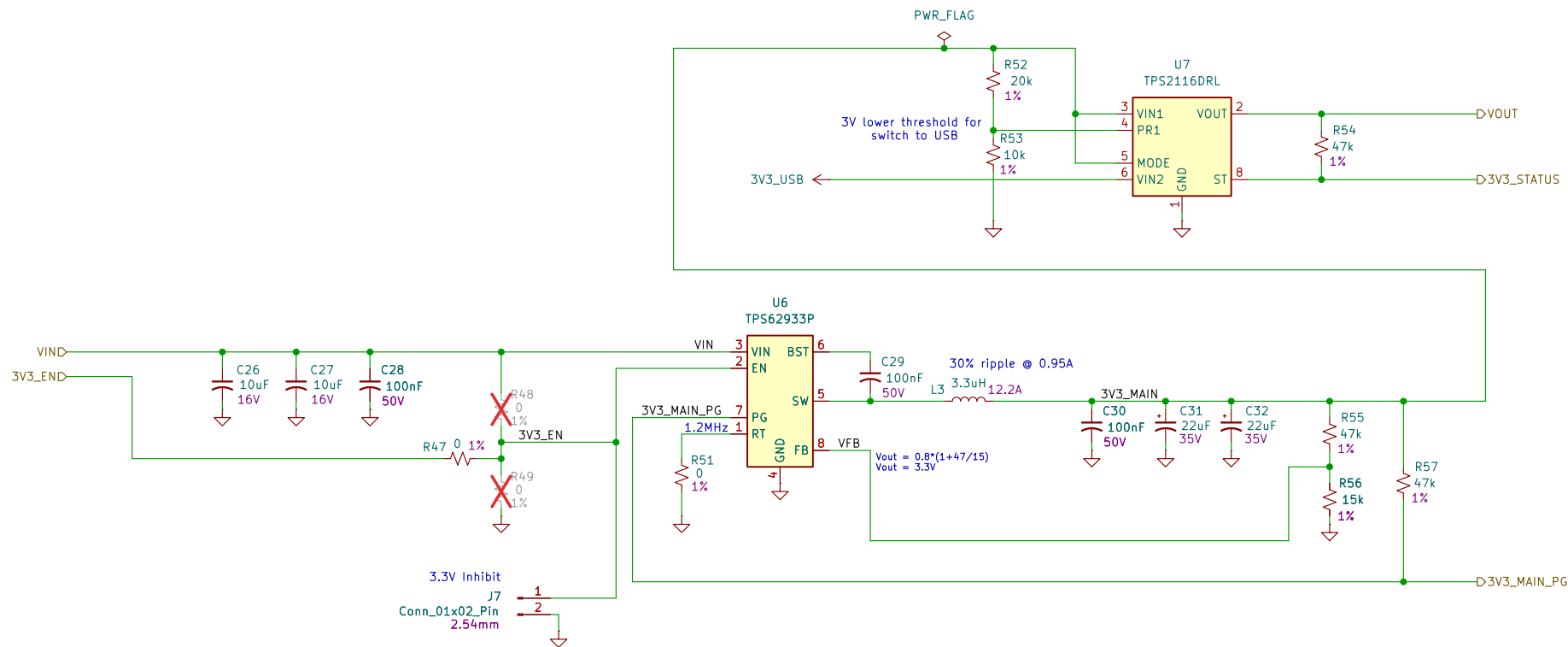
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Size: A4 Date: 2026-08-27

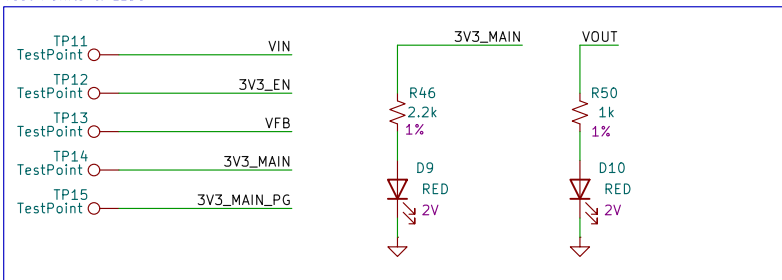
KiCad E.D.A. 10.0.5

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Test Points & LEDs



Sheet: /top-level/power_distribution/3V3_DIGITAL/
File: 3V3_digital.kicad_sch

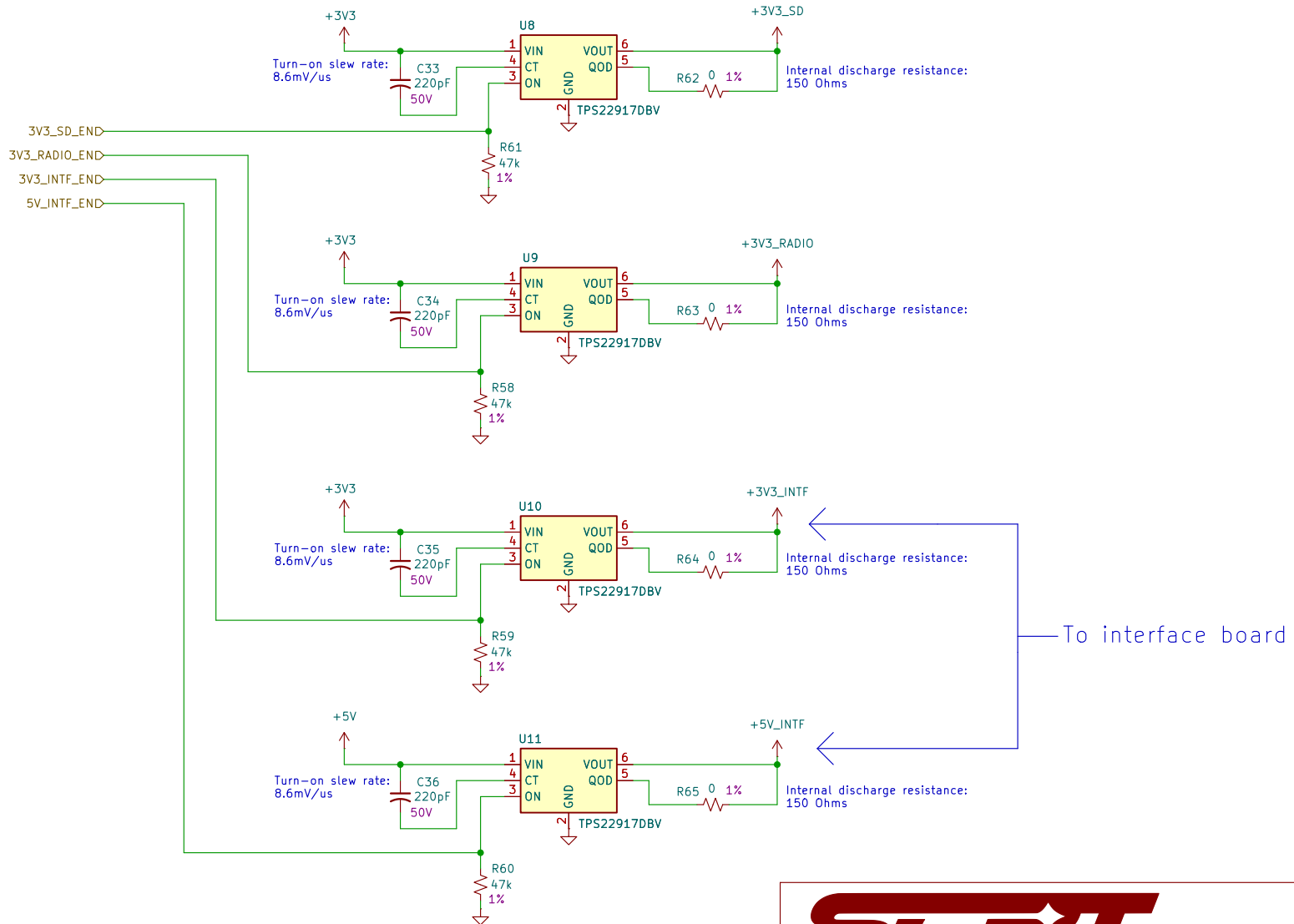
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Size: A4 Date: 2026-08-27

KiCad E.D.A. 10.0.5

Rev: A

Id: 9/27



Sheet: /top-level/power_distribution/load_switches/
File: load_switches.kicad_sch

Title: Load Switches

Size: A4

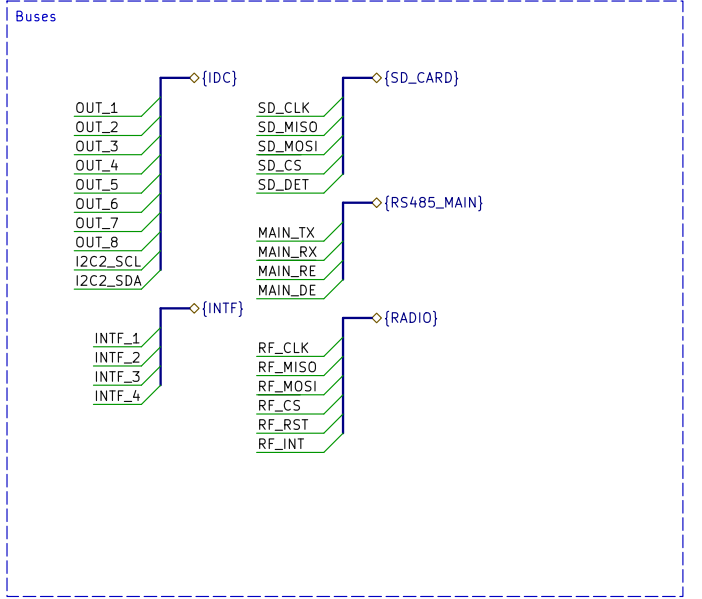
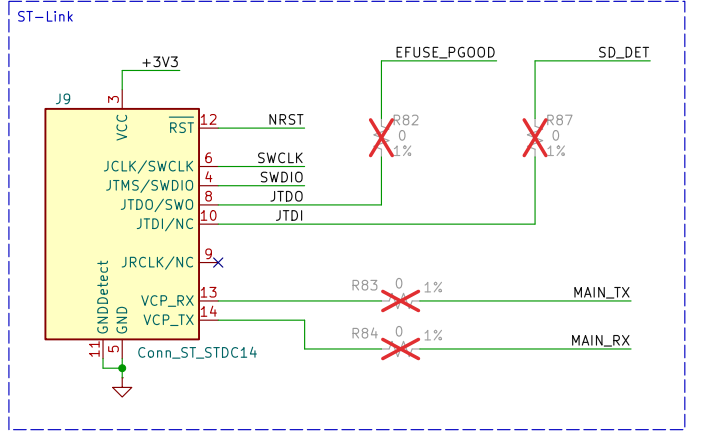
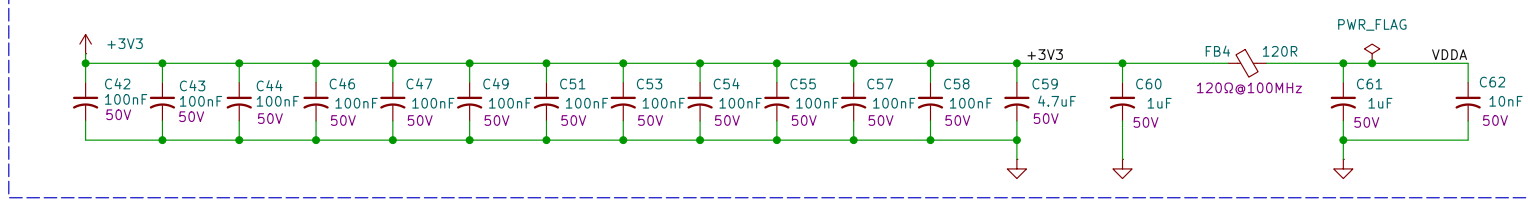
Date: 2026-08-27

Rev: A

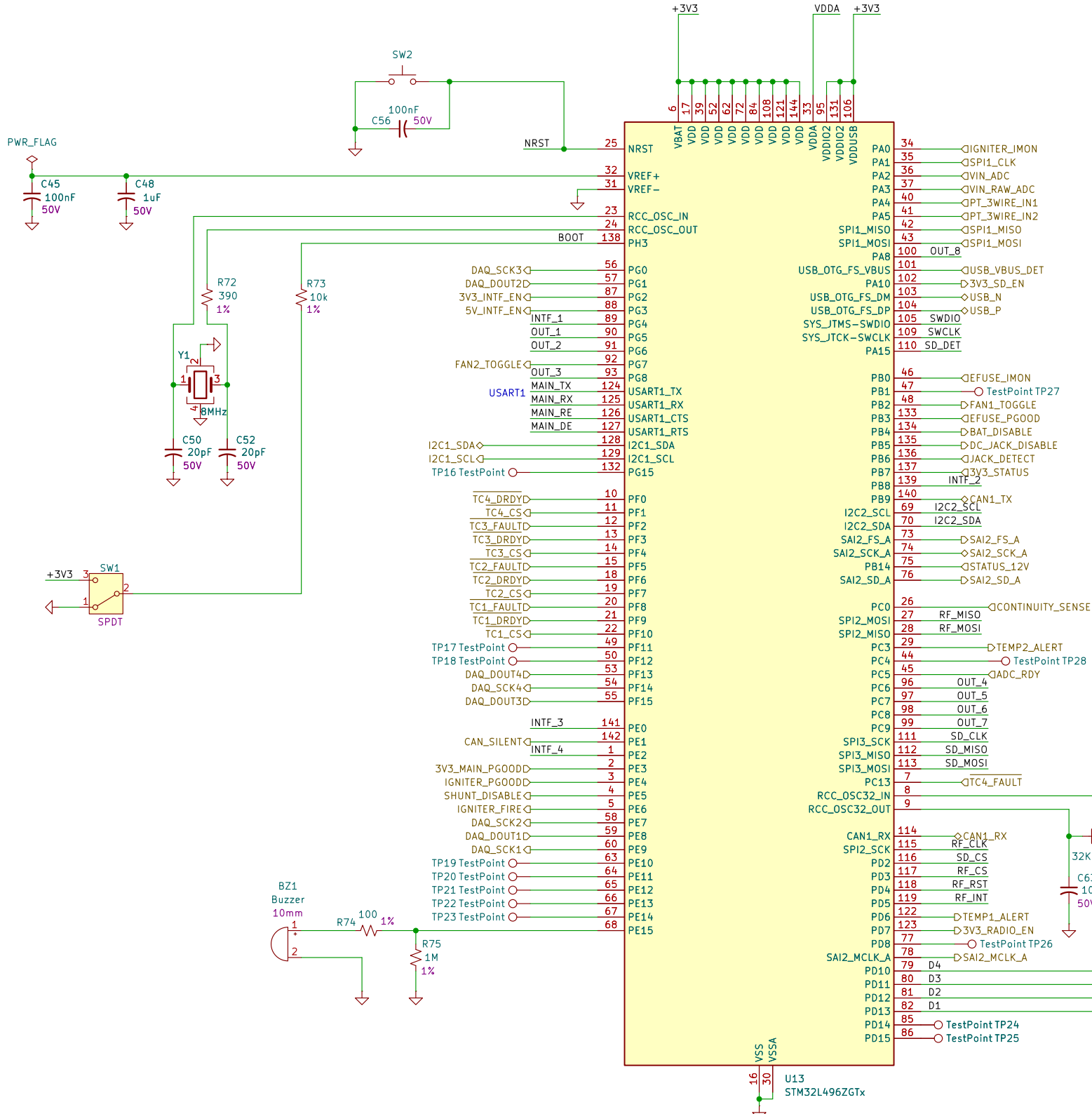
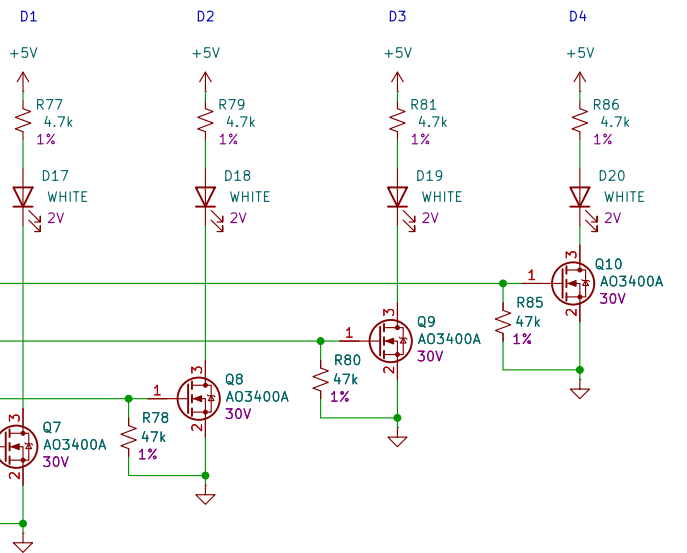
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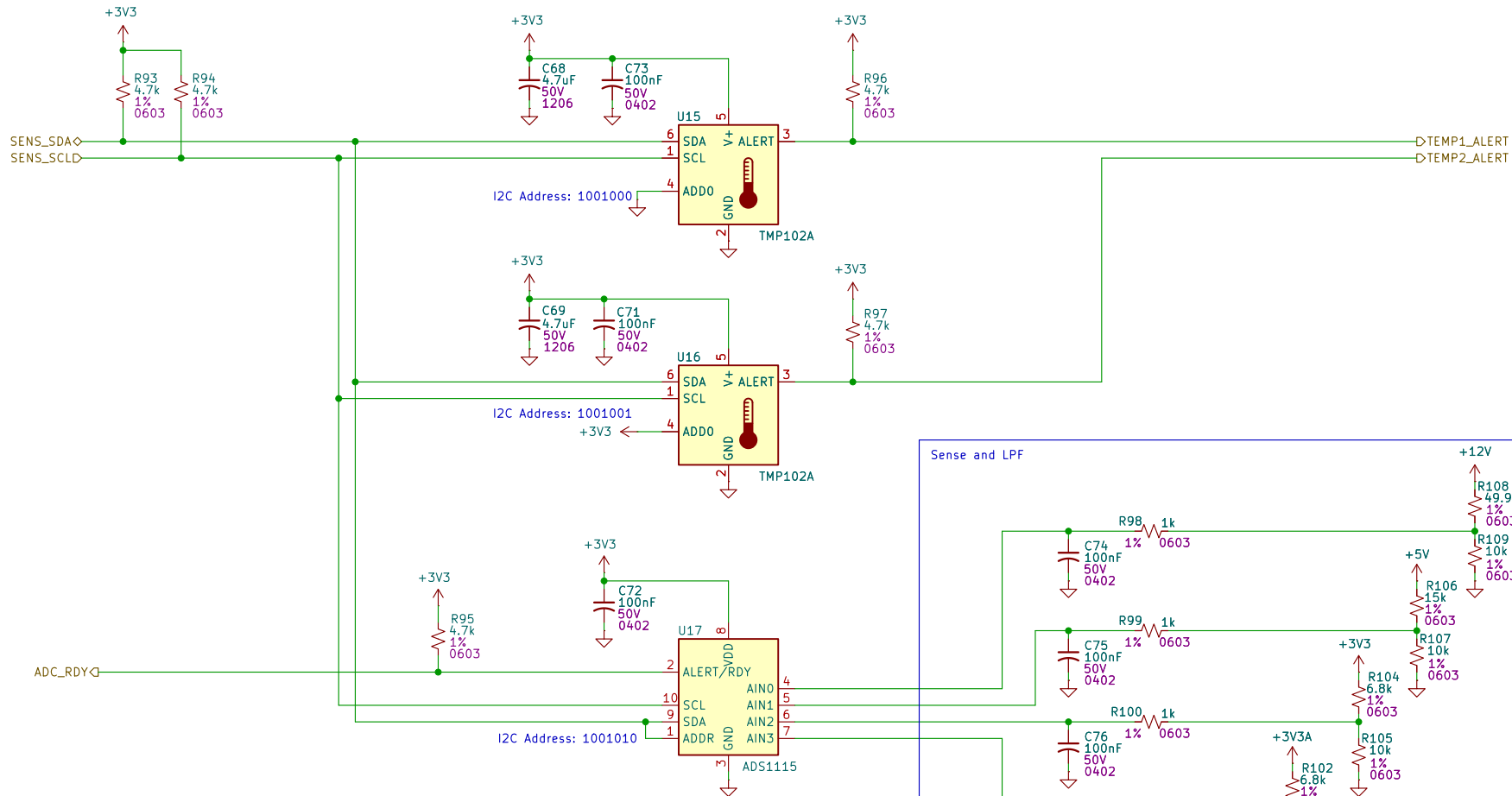
Id: 10/27

Supply & Analog Supply Filtering



Debugging and Status LEDs





Sheet: /top-level/pcb_sensors/
File: pcb_sensors.kicad_sch

Title: PCB Sensors

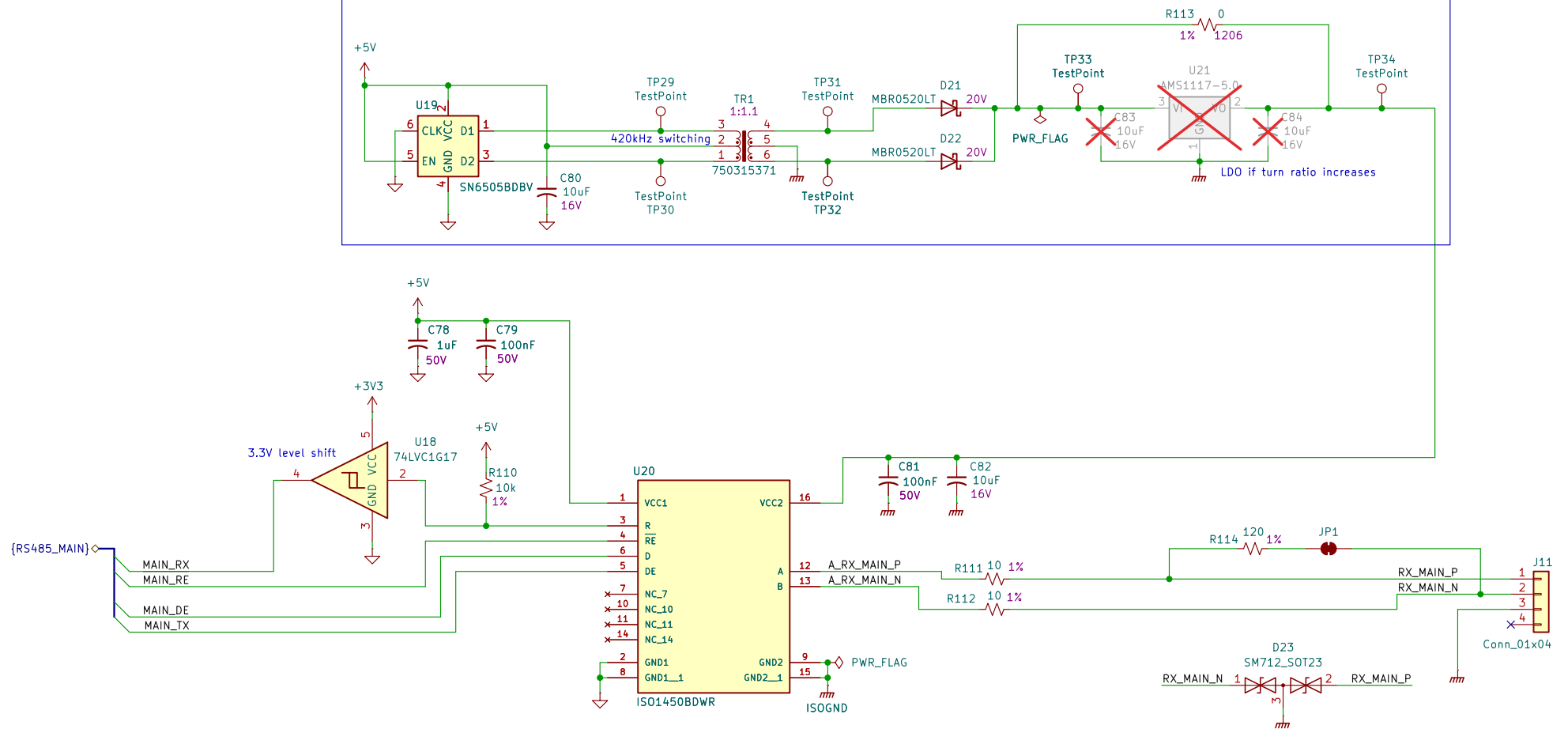
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Rev: A

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Galvanic Power Isolation

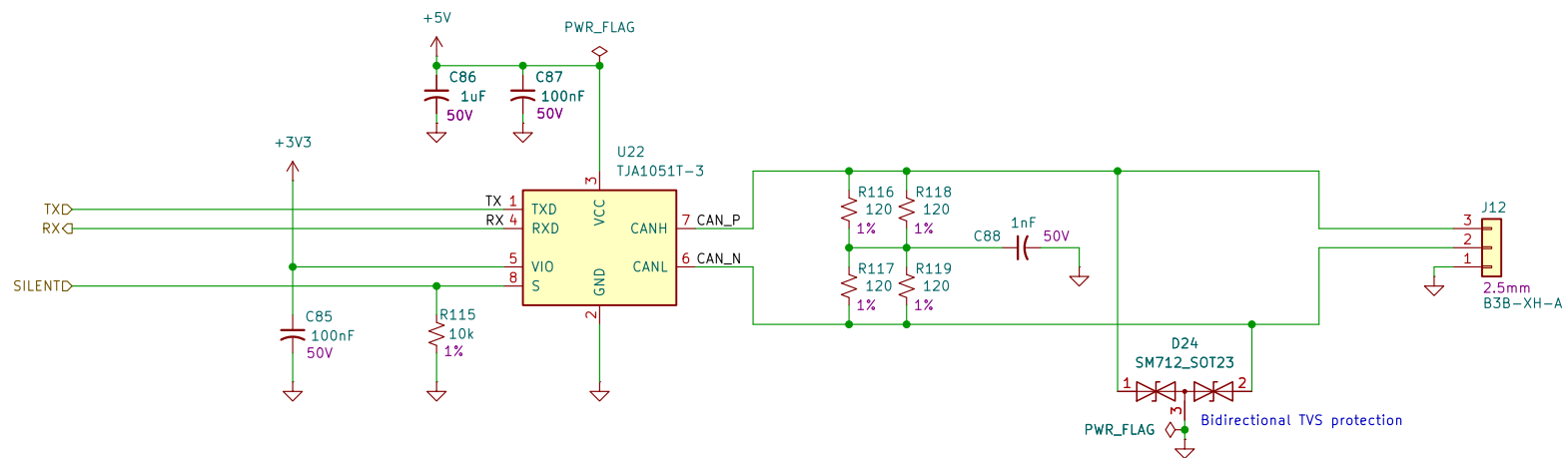


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Title: RS-485

Size: A4 Date: 2026-08-27
KiCad E.D.A. 10.0.5

Rev: A
Id: 13/27



Sheet: /top-level/can_bus/
File: can_bus.kicad_sch

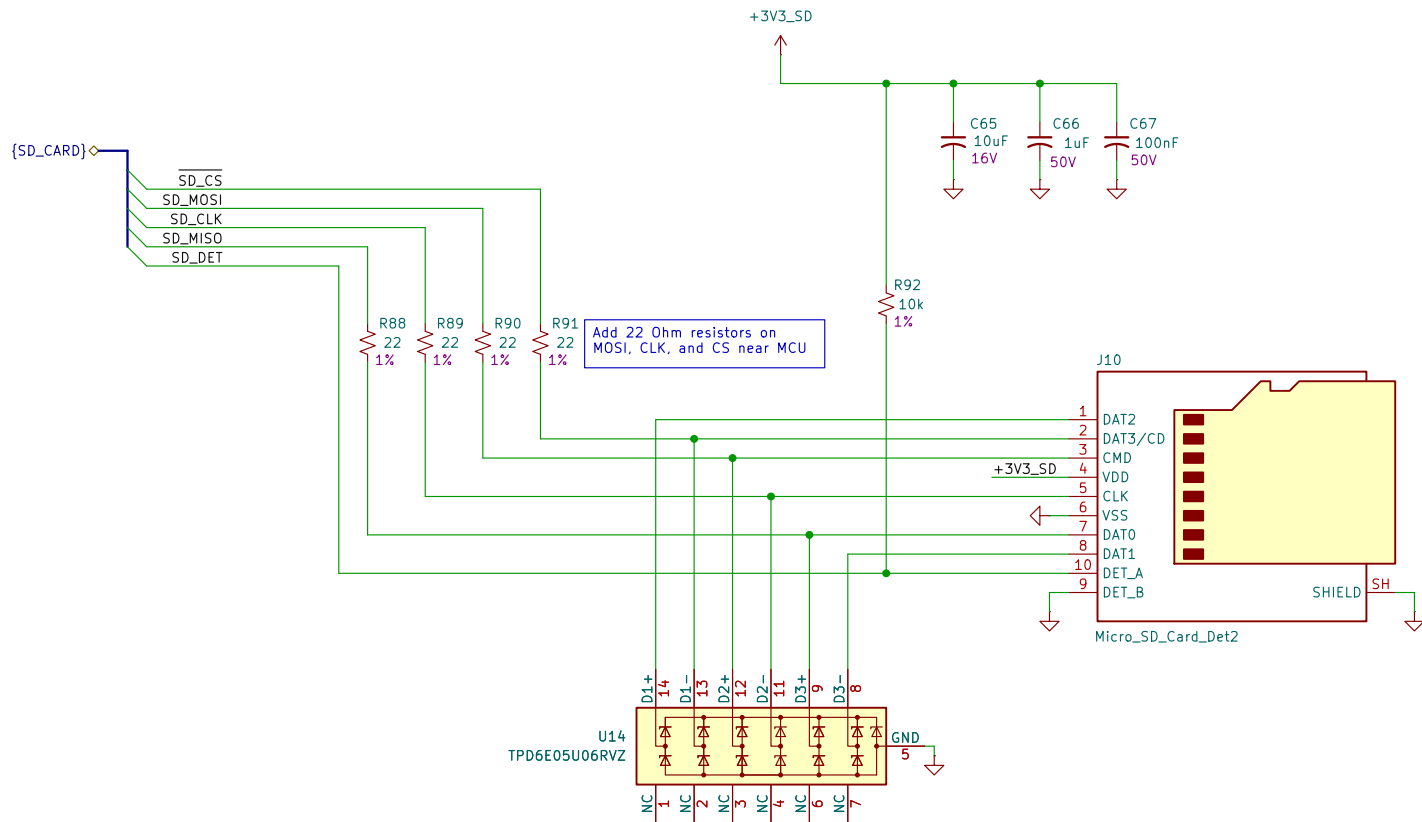
Title: CAN Bus

Size: A4 Date: 2026-08-27

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Rev: A

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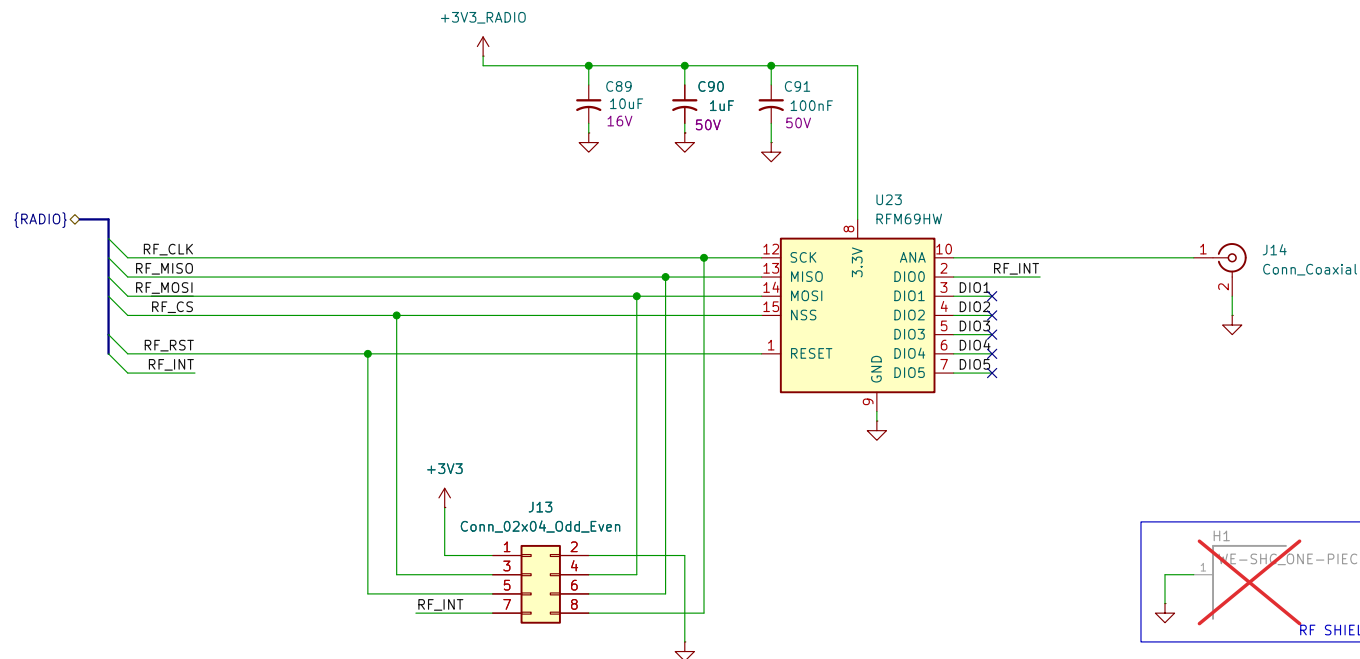
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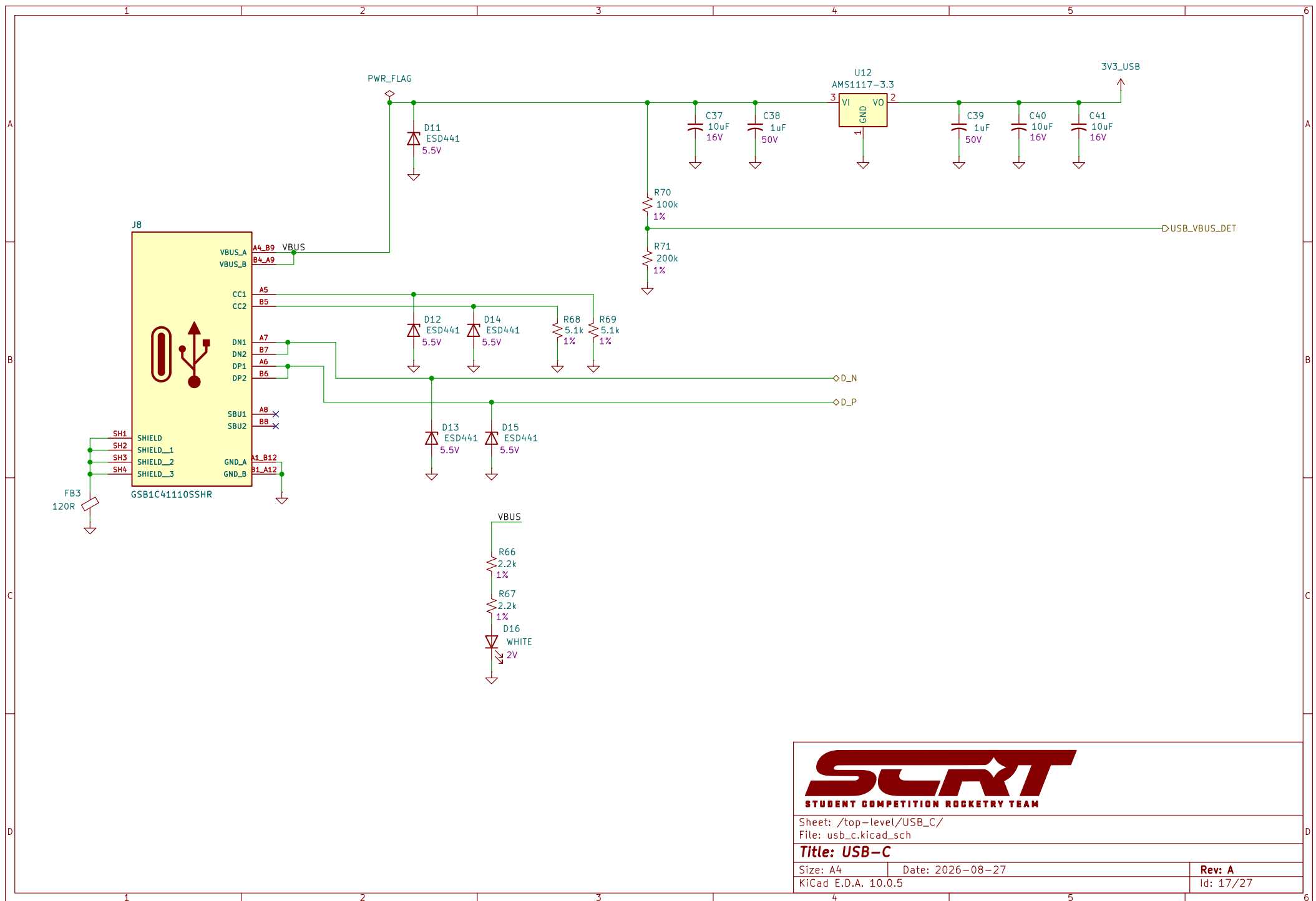
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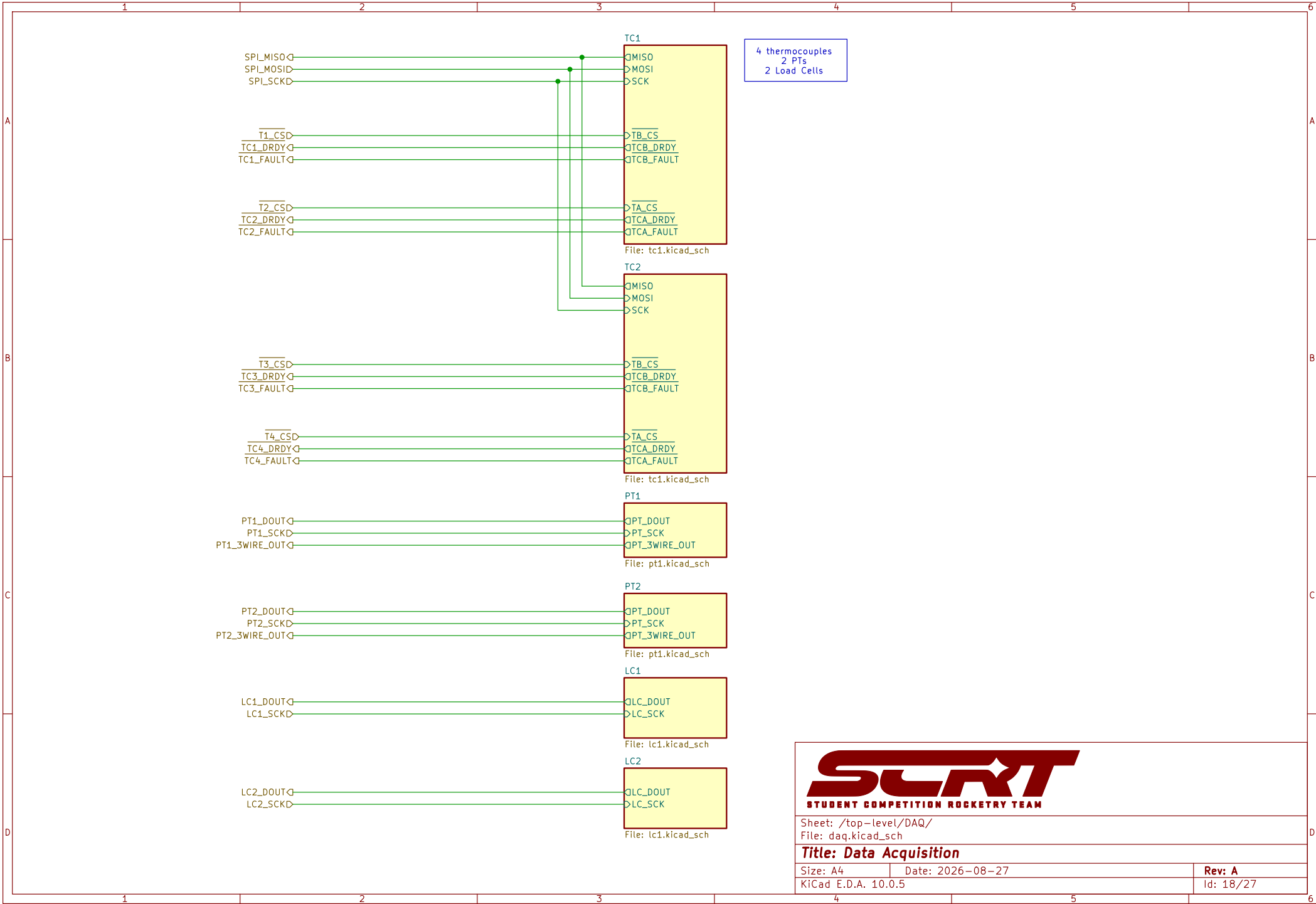
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Rev: A

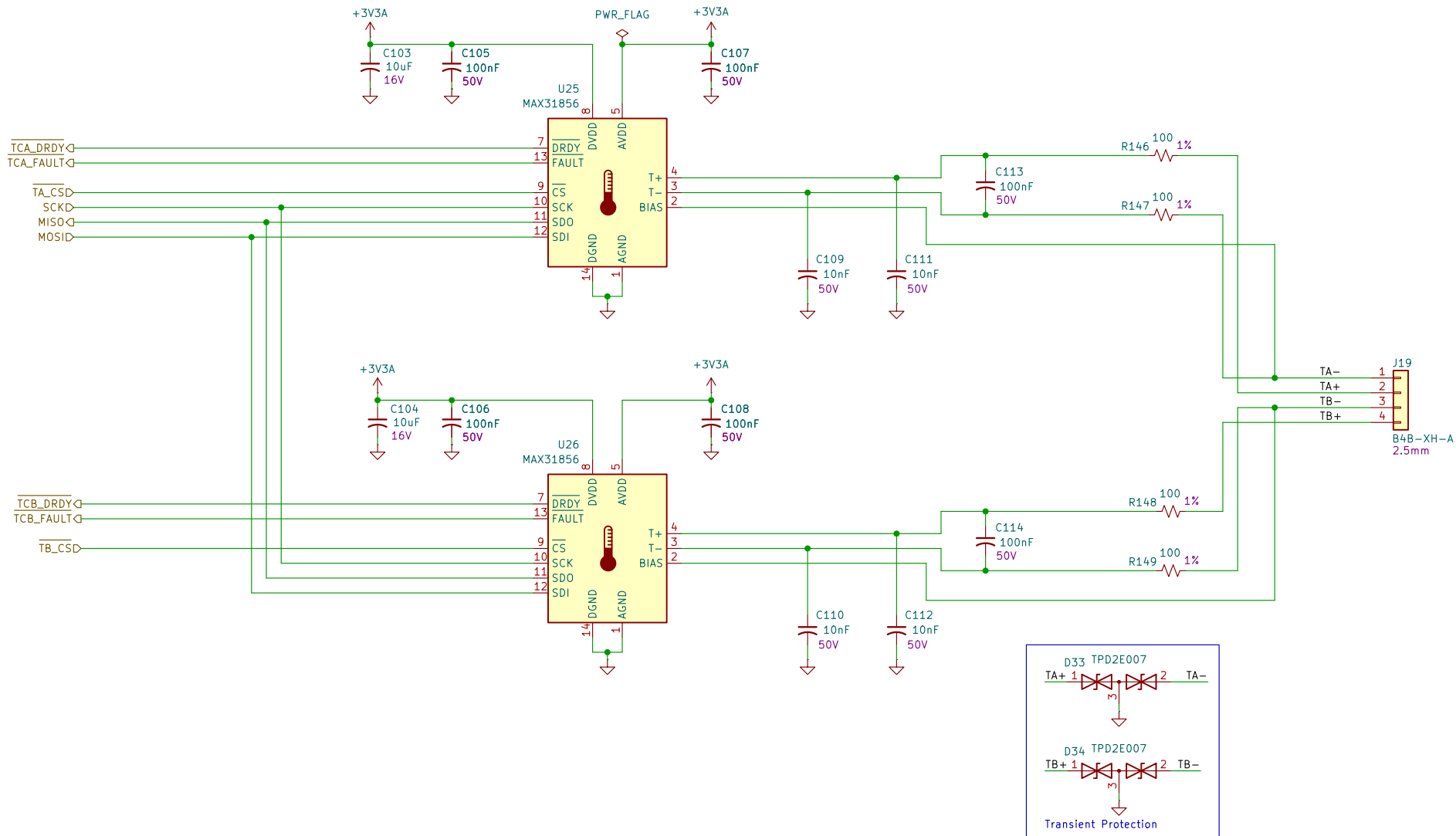
Id: 15/27







Sheet: /top-level/DAQ/ File: daq.kicad_sch		
Title: Data Acquisition		
Size: A4	Date: 2026-08-27	Rev: A
KiCad E.D.A. 10.0.5		Id: 18/27



Sheet: /top-level/DAQ/TC1/
File: tc1.kicad_sch

Title: Thermocouples 3 & 4

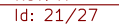
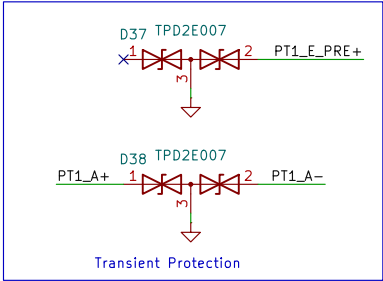
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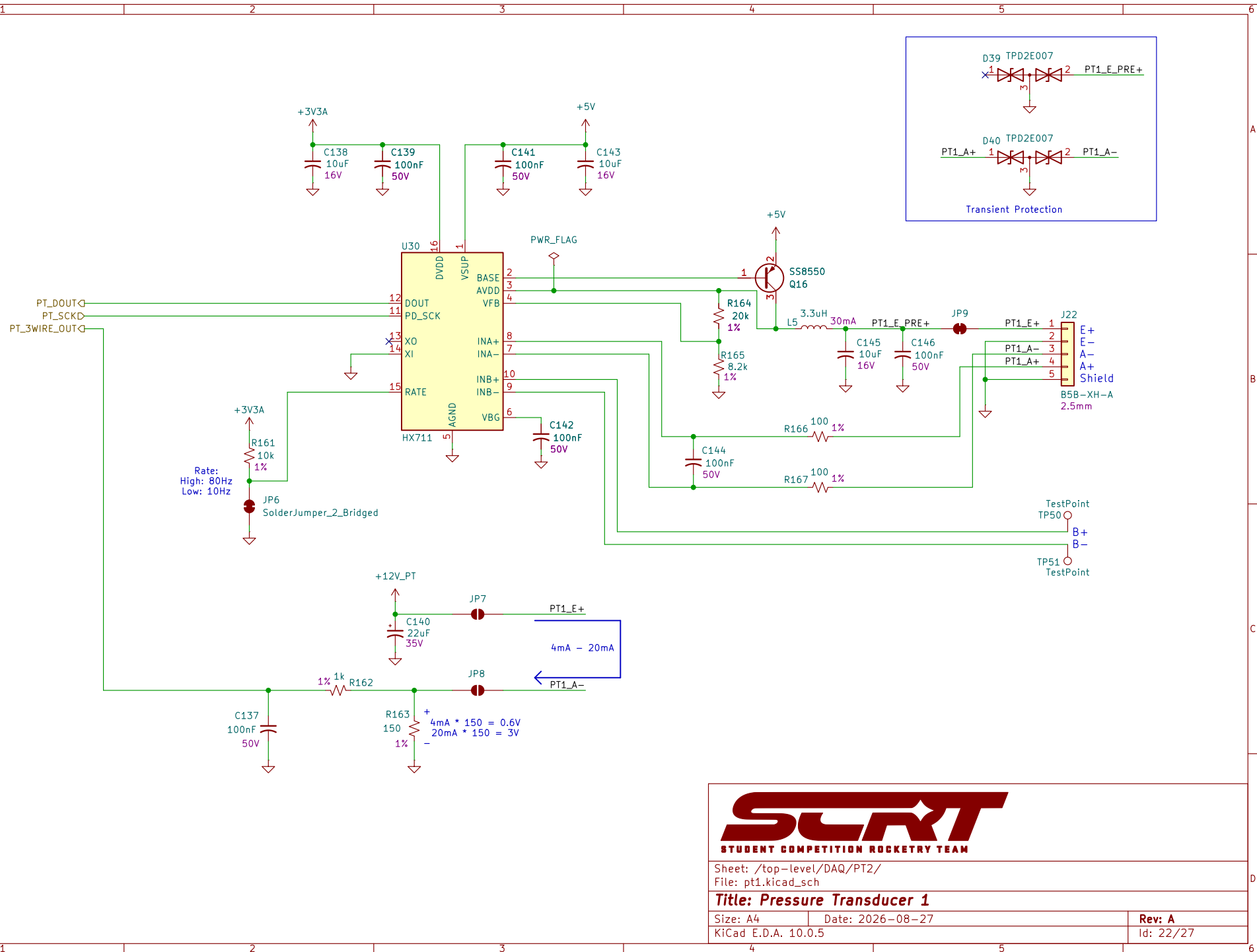
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Sheet: /top-level/DAQ/PT2/
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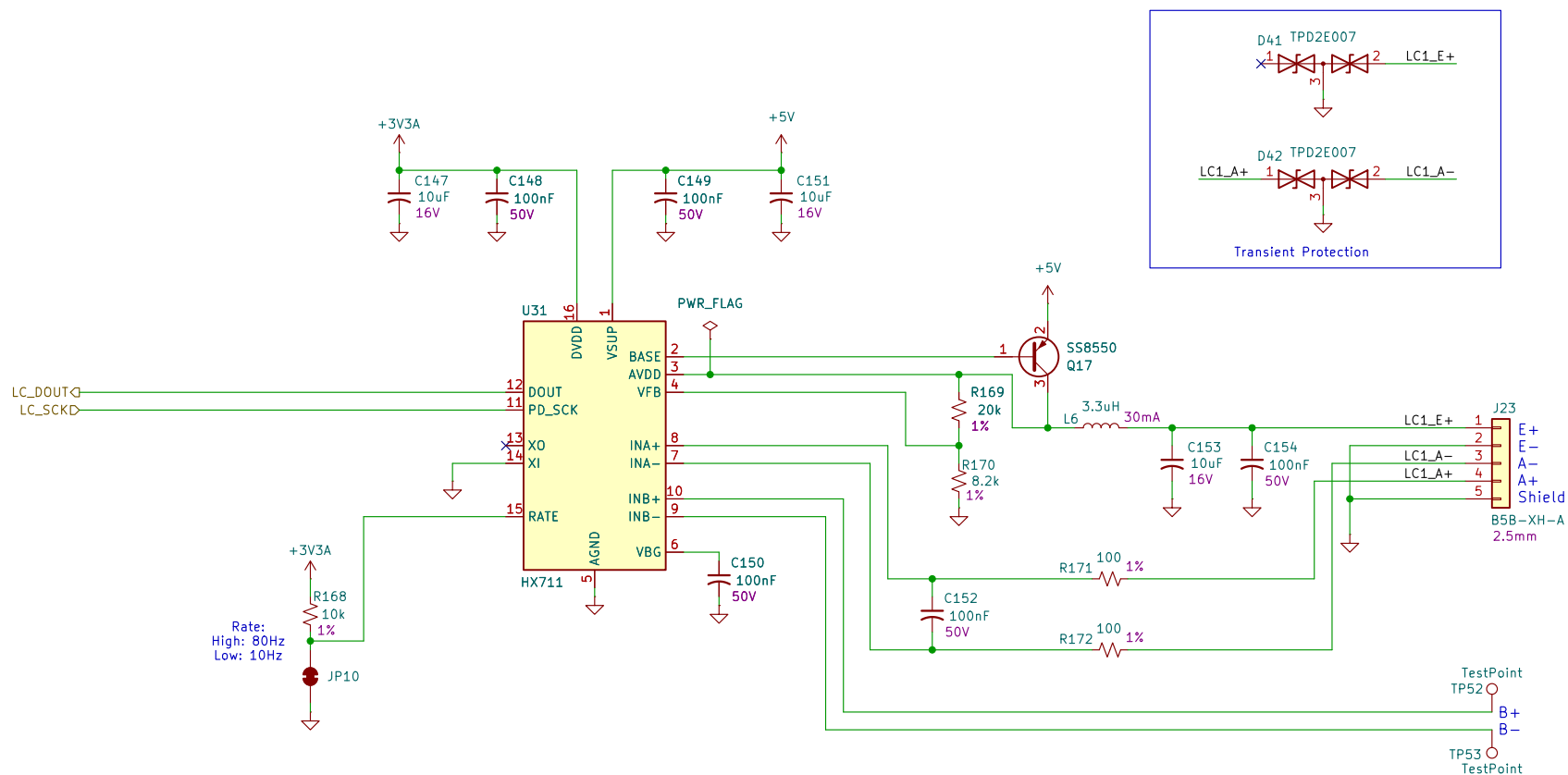
Title: Pressure Transducer 1

Size: A4 Date: 2026-08-27

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Sheet: /top-level/DAQ/LC1/
File: lc1.kicad_sch

Title: Load Cell 1

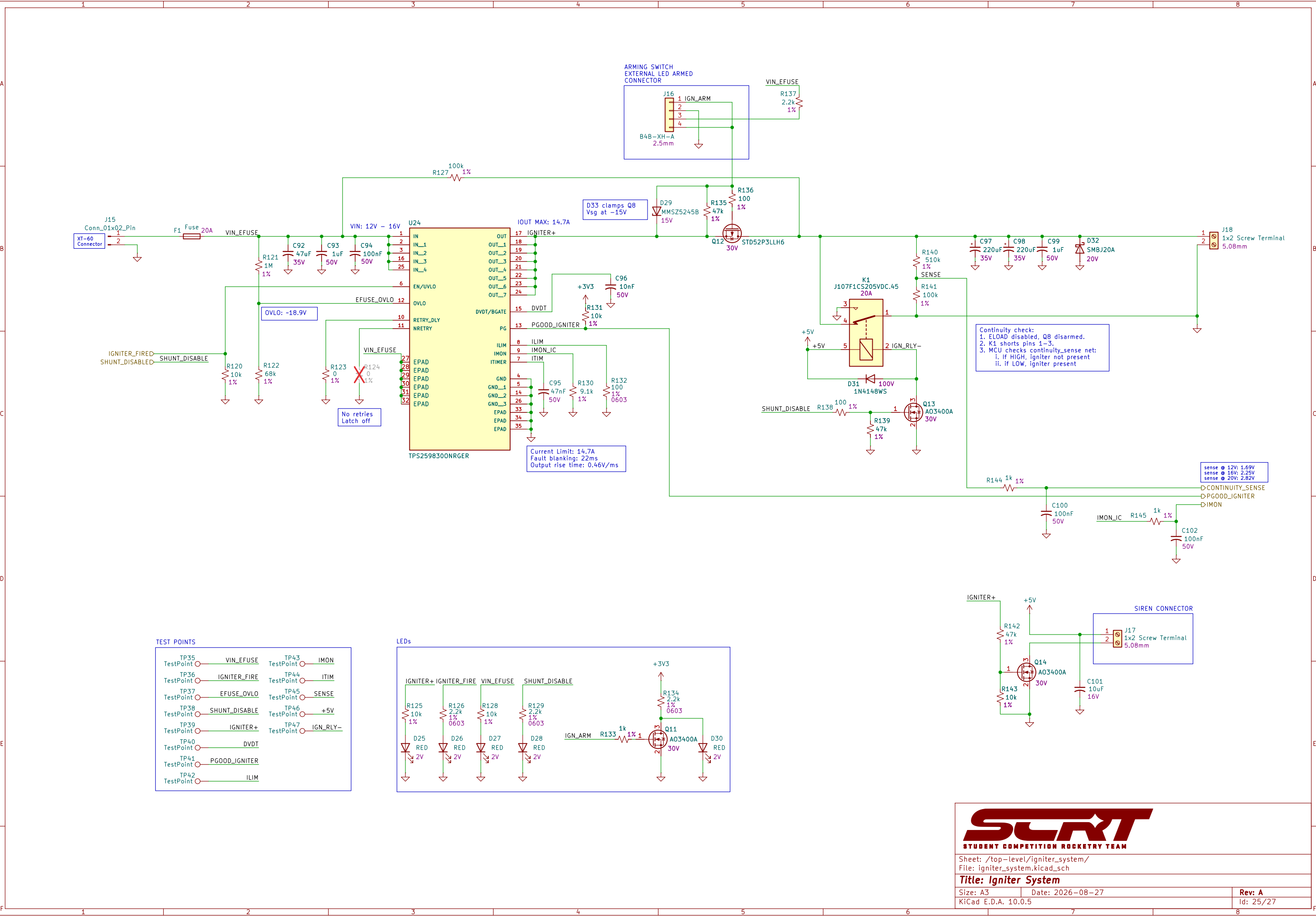
Size: A4 Date: 2026-08-27

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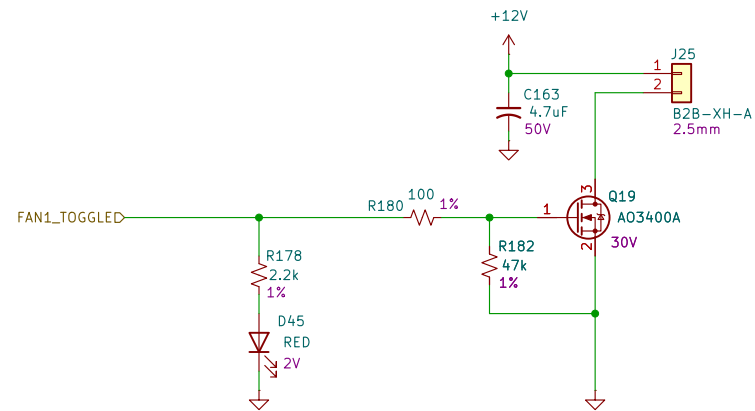
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File: igniter_system.kicad_sch

Title: Igniter System

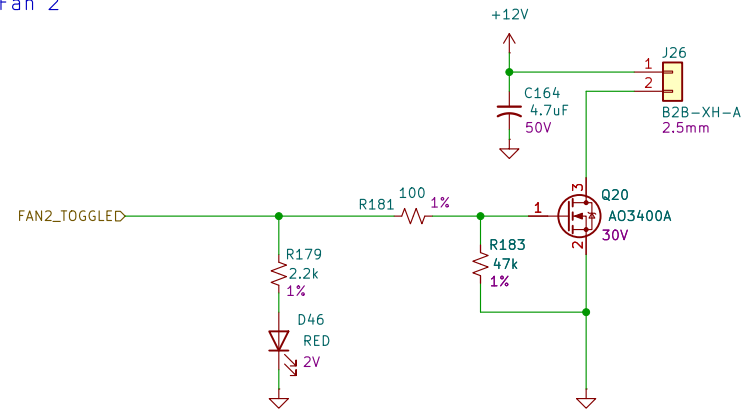
Size: A3 Date: 2026-08-27
KiCad E.D.A. 10.0.5

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Fan 1



Fan 2

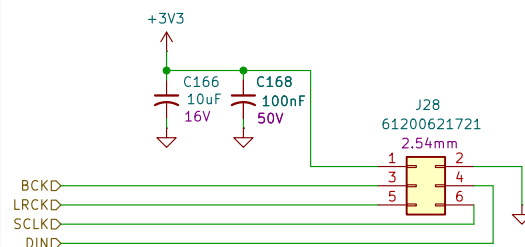


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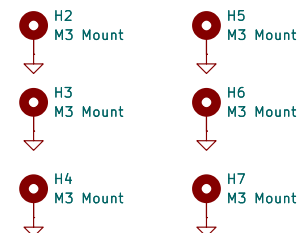
Title: Fans

Size: A4	Date: 2026-08-27	Rev: A
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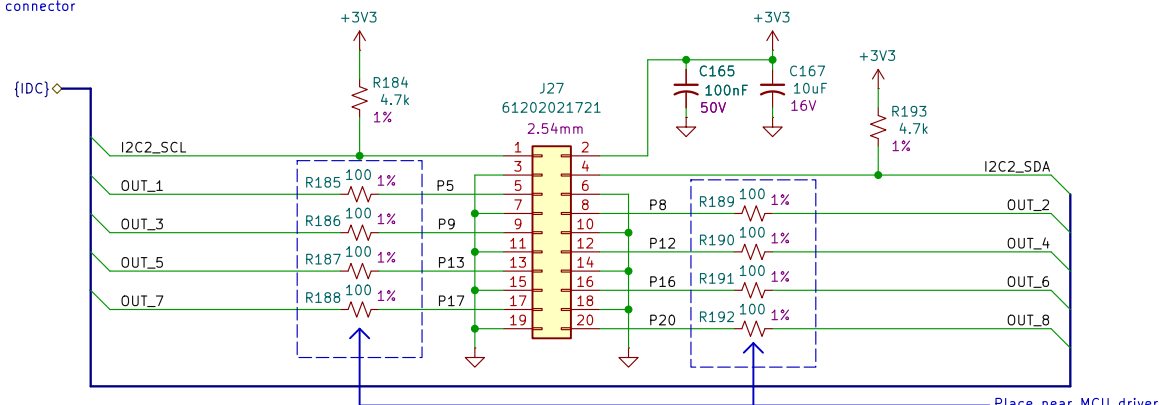
Audio DAC connector



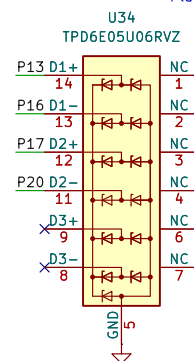
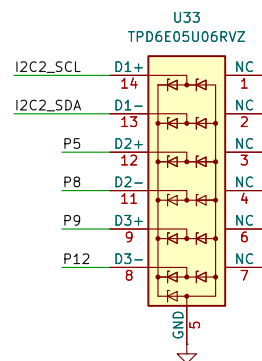
MOUNT HOLES



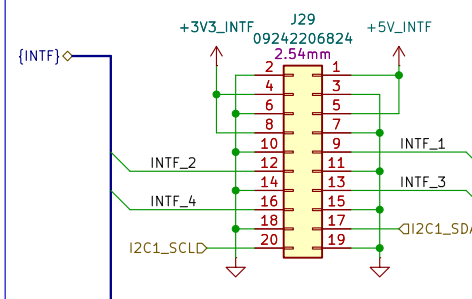
Valve connector



Place near MCU driver



Backplane interface connector



Sheet: /top-level/valve & audio connectors/
File: misc.kicad_sch

Title: MISC

Size: A4 Date: 2026-08-27

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